1 Expand and Simplify $(x+2)(x+4)(x+1)$

$(x+2)(x^2+x+4x+4)$

$(x+2)(x^2+5x+4)$

$x^3+5x^2+4x+2x^2+10x+8$

$x^3+7x^2+14x+8$

$x^3+7x^2+14x+8$

(Total for question 1 is 3 marks)

2 Expand and Simplify $(x-3)(x+5)(x-2)$

$(x-3)(x^2-2x+5x-10)$

$(x-3)(x^2+3x-10)$

$x^3+3x^2-10x-3x^2-9x+30$

$x^3-19x+30$

$x^3-19x+30$

(Total for question 2 is 3 marks)

3 Expand and Simplify $(x + 2)(x + 1)(x + 5)$

$(x+2)(x^2+5x+x+5)$

$(x+2)(x^2+6x+5)$

$x^3+6x^2+5x+2x^2+12x+10$

$x^3+8x^2+17x+10$

$x^3+8x^2+17x+10$

(Total for question 3 is 3 marks)

4 Expand and Simplify $(x + 4)(x + 5)(x - 4)$

$(x+4)(x^2-4x+5x-20)$

$(x+4)(x^2+x-20)$

$x^3+x^2-20x+4x^2+4x-80$

$x^3+5x^2-16x-80$

$x^3+5x^2-16x-80$

(Total for question 4 is 3 marks)

5 Expand and Simplify $(x + 3)(x - 1)^2$

$(x+3)(x-1)(x-1)$

$(x+3)(x^2 - x - x + 1)$

$(x+3)(x^2 - 2x + 1)$

$x^3 - 2x^2 + x + 3x^2 - 6x + 3$

$x^3 + x^2 - 5x + 3$

$x^3 + x^2 - 5x + 3$

(Total for question 5 is 3 marks)

6 Expand and Simplify $(x + 5)(x - 3)(2x - 1)$

$(x+5)(2x^2 - x - 6x + 3)$

$(x+5)(2x^2 - 7x + 3)$

$2x^3 - 7x^2 + 3x + 10x^2 - 35x + 15$

$2x^3 + 3x^2 - 32x + 15$

$2x^3 + 3x^2 - 32x + 15$

(Total for question 6 is 3 marks)

7 Expand and Simplify $(2x + 1)(x + 2)(x + 3)$

$(2x+1)(x^2+3x+2x+6)$

$(2x+1)(x^2+5x+6)$

$2x^3+10x^2+12x+x^2+5x+6$

$2x^3+11x^2+17x+6$

$2x^3+11x^2+17x+6$

(Total for question 7 is 3 marks)

8 Expand and Simplify $(2x - 3)(x - 2)(3x - 1)$

$(2x-3)(3x^2-x-6x+2)$

$(2x-3)(3x^2-7x+2)$

$6x^3-14x^2+4x-9x^2+21x-6$

$6x^3-23x^2+25x-6$

$6x^3-23x^2+25x-6$

(Total for question 8 is 3 marks)

9 Expand and Simplify $(x-2)(3x+2)(x+5)$

$(x-2)(3x^2+15x+2x+10)$

$(x-2)(3x^2+17x+10)$

$3x^3+17x^2+10x-6x^2-34x-20$

$3x^3+11x^2-24x-20$

$3x^3+11x^2-24x-20$

(Total for question 9 is 3 marks)

10 Expand and Simplify $(3x+1)(x+2)(x-4)$

$(3x+1)(x^2-4x+2x-8)$

$(3x+1)(x^2-2x-8)$

$3x^3-6x^2-24x+x^2-2x-8$

$3x^3-5x^2-26x-8$

$3x^3-5x^2-26x-8$

(Total for question 10 is 3 marks)

11 Show that $(2x + 3)(5x + 2)(x - 5) = 10x^3 - 31x^2 - 89x - 30$

for all values of x.

$(2x+3)(5x^2 - 25x + 2x - 10)$

$(2x+3)(5x^2 - 23x - 10)$

$10x^3 - 46x^2 - 20x + 15x^2 - 69x - 30$

$\underline{10x^3 - 31x^2 - 89x - 30}$

.....

(Total for question 11 is 3 marks)

12 Show that $(2x - 1)(3x + 2)^2 = 18x^3 + 15x^2 - 4x - 4$

for all values of x.

$(2x-1)(3x+2)(3x+2)$

$(2x-1)(9x^2 + 6x + 6x + 4)$

$(2x-1)(9x^2 + 12x + 4)$

$18x^3 + 24x^2 + 8x - 9x^2 - 12x - 4$

$\underline{18x^3 + 15x^2 - 4x - 4}$

.....

(Total for question 12 is 3 marks)

13 Show that $(3x - 1)(4x + 3)(x - 9) = 12x^3 - 103x^2 - 48x + 27$

for all values of x.

$(3x - 1)(4x^2 - 36x + 3x - 27)$

$(3x - 1)(4x^2 - 33x - 27)$

$12x^3 - 99x^2 - 81x - 4x^2 + 33x + 27$

$12x^3 - 103x^2 - 48x + 27$

.....

(Total for question 13 is 3 marks)

14 Show that $(5x - 4)(3x + 1)(2x - 7) = 30x^3 - 119x^2 + 41x + 28$

for all values of x.

$(5x - 4)(6x^2 - 21x + 2x - 7)$

$(5x - 4)(6x^2 - 19x - 7)$

$30x^3 - 95x^2 - 35x - 24x^2 + 76x + 28$

$30x^3 - 119x^2 + 41x + 28$

.....

(Total for question 14 is 3 marks)